Milestone 03 Testing Summary Report

CP476B - Internet Computing | Group 6 | Final verification: July 23, 2026

Application	BudgetBridge Personal Expense Tracker	Final mode	Node.js + MySQL
Automated result	12 passed, 0 failed	Database result	Persistence verified
Target environment	Node.js 18+ / MySQL 8.x / modern browser	Demo length	Approximately 9:22

Purpose

This report records the final automated and manual verification evidence for BudgetBridge. It covers the full-stack workflow, database persistence, input validation, security controls, the final MySQL report defect and resolution, known limitations, and reproduction instructions.

1. Test Strategy and Evidence

Automated unit/security	Validation, password hashing, session-token behavior	Node.js built-in test runner
Automated API integration	Authentication, CRUD, reports, settings, filters, isolation	12 tests passed; 0 failed
Manual MySQL smoke test	Schema, server startup, persistence, dashboard, categories, reports	Workbench and browser captures
Static/project review	Repository structure, README, environment notes, documentation	Final GitHub package

Final test command

npm test produced 12 passed, 0 failed, 0 skipped. The captured run covered the full authenticated workflow and security/validation checks.

```
C:\Users\theso\Downloads\budgetbridge-expense-tracker-main (1)\budgetbridge-expense-tracker-main>npm test

> budgetbridge-expense-tracker@3.0.0 test
> node --test tests/*.test.js

✔ full authenticated CRUD workflow and reports (331.8531ms)
✔ users cannot access another users transaction (247.4633ms)
✔ custom category CRUD and category protection rules (136.5486ms)
✔ filters and search return only matching transactions (135.4238ms)
✔ profile and password update invalidate old session (482.5152ms)
✔ invalid data and duplicate accounts return clear errors (300.3881ms)
✔ password hash verifies only the correct password (318.7834ms)
✔ session tokens are random and stored as hashes (2.1997ms)
✔ registration normalizes email and accepts strong password (2.983ms)
✔ weak password is rejected (1.6148ms)
✔ invalid transaction amount is rejected (0.8073ms)
✔ invalid date range is rejected (3.1427ms)
ℹ tests 12
ℹ suites 0
ℹ pass 12
ℹ fail 0
ℹ cancelled 0
ℹ skipped 0
ℹ todo 0
ℹ duration_ms 1915.186
```

Final automated test output captured on July 23, 2026.

2. Automated Test Results

AT-01	Full authenticated CRUD workflow and reports	PASS
AT-02	Block access to another user’s transaction	PASS
AT-03	Custom category CRUD and protection rules	PASS
AT-04	Filters and search return only matching transactions	PASS
AT-05	Profile/password update invalidates old session	PASS
AT-06	Invalid data and duplicate accounts return clean errors	PASS
AT-07	Password hash verifies only the correct password	PASS
AT-08	Session tokens are random and stored as hashes	PASS
AT-09	Registration normalizes email and accepts strong password	PASS
AT-10	Weak password is rejected	PASS
AT-11	Invalid transaction amount is rejected	PASS
AT-12	Invalid date range is rejected	PASS

Security and data-integrity controls verified

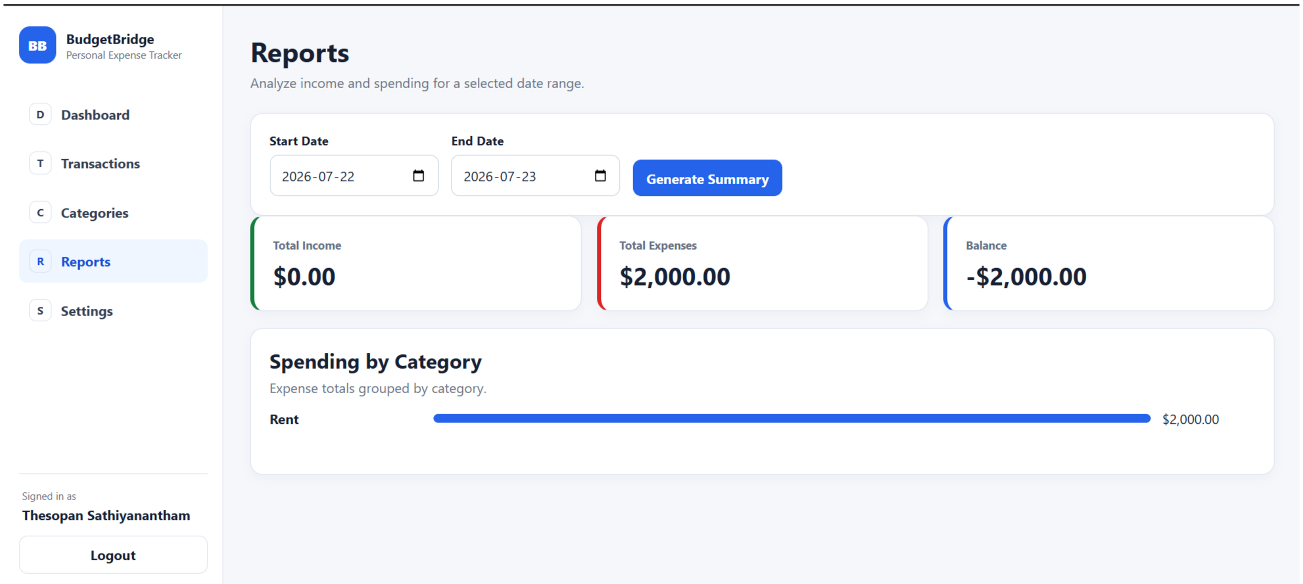
• Passwords are hashed with scrypt and compared using timing-safe logic.
• Raw session tokens are not stored in MySQL; only SHA-256 token hashes are retained.
• Protected endpoints require an authenticated session.
• Transaction and category queries include the authenticated user ID.
• Amounts, dates, category IDs, names, email addresses, searches, and passwords are validated server-side.
• User-controlled SQL values are parameterized; the recent-record limit is parsed and constrained before interpolation.
• Database primary keys, foreign keys, unique rules, checks, and indexes preserve integrity.

3. Final Manual MySQL Verification

MT-01	Execute schema.sql	Five tables created in budgetbridge_db	PASS
MT-02	Start server in MySQL mode	Server reported Database mode: mysql	PASS
MT-03	Register/login and create default categories	Account and categories available	PASS
MT-04	Logout/login persistence check	Transactions remained stored	PASS
MT-05	Add income and expense records	Dashboard totals updated	PASS
MT-06	Open and edit a transaction	Edit form loaded stored record	PASS
MT-07	Create a custom category	Custom category appeared with edit/delete controls	PASS
MT-08	Generate date-range report	Totals and category aggregation displayed	PASS after fix
MT-09	View account settings	Profile and password controls loaded	PASS
MT-10	Run final automated suite	12 passed, 0 failed	PASS

4. Defect Record and Retest

Defect	Reports returned a server error in MySQL mode.
Observed database error	ER_WRONG_ARGUMENTS: Incorrect arguments to mysqld_stmt_execute.
Root cause	The recent-transactions query used a prepared placeholder in LIMIT ?. The local MySQL/driver combination rejected that argument.
Correction	The limit is now parsed as an integer, restricted to 1-100, and appended to the SQL statement. All user-controlled search/filter values remain parameterized.
Retest result	PASS - report totals and spending-by-category displayed correctly



Report page after the MySQL LIMIT fix and successful retest.

5. Known Limitations and Final Assessment

- The application is designed for local course demonstration rather than public production hosting.
- Amounts are displayed in Canadian dollars only.
- Bank synchronization, email password reset, multi-factor authentication, and forecasting are outside the approved scope.
- The final video is a polished screenshot-driven walkthrough using actual application, MySQL, GitHub, and test evidence rather than one continuous live-click recording.

Final assessment: BudgetBridge satisfies the required full-stack workflow, database-backed CRUD, validation, basic security hygiene, testing evidence, clean-machine documentation, project tracking, presentation artifact, and final demo requirements.

How to reproduce

- Install Node.js 18+ and MySQL 8.x.
- Run database/schema.sql.
- Copy .env.example to .env and enter local MySQL credentials.
- Run npm install, then npm start.
- Open http://localhost:3000 and test the workflow.
- Stop the server and run npm test.